

ActInf Textbook Group ~ Cohort 6 ~ Session 3

(Chapter 1, Part 2) ~ 2/12/2024

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all right welcome everyone it's cohort 6 and our second discussion of chapter one so Andrew and Ali thanks for facilitating just take it from here wherever you want to pick up Andrew would you mind taking the lead on this one sure yeah so I know um yeah absolutely so welcome cohort 6 this is week two that we're spending on chapter one of the textbook um I'm not sure what was discussed at the last meeting but I thought it might be worthwhile for some of the folks who are unable to attend the morning meetings to maybe just quickly move through um the sections of textbook um just to quickly get a bit of a recap or overview um yeah the overview we begin with a nice brief introduction um describing what's going to be happening in this chapter we already get off the bat that active inference is a normative perspective it's starting from first principles um move on to section 1.2 get some quick explanations regarding the relationships between things like actions and sensory observations specifically that living organisms can only maintain their bodily Integrity by exerting adaptive control over the action perception Loop this means acting to solicit sensory observations that either correspond to desired outcomes or goals such as the sensations that accompany secure nutrients and shelter for simple organisms or friends in jobs for more complex ones or help in making sense of world um this is carried out through there's a rather recursive nature to this cycle of action uh perception Loops um also whenever we move on to looking at more complex organisms there's the assumption that organisms can increasingly use like more complex adaptive strategies to adjust to their environment um section 1.3 still more on um first principles get an introduction to this uh this kind of I don't want to call it ad hoc because I think these conversations have been had before but it's a a little light-hearted I suppose this meets versus scruffy's uh division that's main right and so um scruffies being like these are folks who will approach these kinds of problems but maybe from within their particular field or subfield um they don't want to um you know forego any kind of kind of idiocy idiosyncrasies or specificities to their object of study um so maybe you know not so quick to

jump to some kind of broader unifying framework um to analyze things whenever you know you might let go of some of the specific the problem you're looking at um you anthropologists will have disagreements with sociologists and so on about what what defines like a society or or a culture or otherwise um nonetheless meats are those who do seek unification Beyond heterogeneity of brain and mind phenomena and that is the perspective that active in for instincts it does in fact attempt a kind of unified framework at looking at diverse uh scenarios and organisms ranging from human beings and entire cultures uh who are operating at different time scales of the way at fast time scales of like what they're immediately observing and perceiving um all the way up to slower time scales of like planning learning um organizing with a group development of Institutions and so on um section 1.4 gives us a good overview kind of structure of the book um 1.4.1 active inference in theory we get more about surprise minimization um we're also introduced to variational free energy in the next paragraph there um they keep it short in describing why we might look at variational free energy as opposed to surprise which seems like a much more intuitive concept um for one to be surprised at unexpected uh phenomena or observations or incoming um sensory signals um but the the to put it quickly variational free energy is basically an upper bound on surprisal that is done because computationally um we find this issue of intractability in computation when whenever you break down the math you find like oh in order to actually precisely compute surprisal you would need something like an entire history of an organism's like memory every observation it's come across every potential in state it would estimate and you can imagine that the the the the mathematics to that the kind of formula that would be involved in that would be um just untenable from a computational perspective and you can imagine the same thing for an actual human brain um and making snap judgments based on its entire history of what it's learned and so on so variational free energy is is very important in order to overcome this kind of intractability problem everything is somewhat approximate with the goal being to minimize free energy as much as you can such as surprises also minimized as much as possible there will be much more on that in the textbook we're only in chapter one um so stepping forward from that um we do have this nice figure 1.2 which describes um how to reach active inference from two different routes uh the low road and the high road we can say that the low road is kind of the the how of active inference how an organism maintains its own Integrity um just before we started this meeting there were certain uh mentions of autopois and so on like this is the how and and so it it relies heavily upon basis theorem uh we work up to generative models and using bases theorem in order to find

relationships between different variables within the model um and so that's kind of the mechanics it's the how but from The High Road alternatively we get this is where things become a little bit more of aose uh which is that this is what and why organisms are doing what they're doing as far as their behavior goes and attempting to explain like what is happening here um they're minimizing free energy so we begin with the free energy principle right so we so then we can start understand living organisms or other kinds of systems that we attempt to model using these kinds of tools on the one hand they're trying to minimize free energy on the other hand how are they doing it they're doing it through something like Basi and statistics mechanics um we'll get much more on on both of those roads to active inference in chapters 2 and three Each of which are dedicated to one route uh one route each following that learn that yourself when we look at the structure of the book as it's described from there um to keep it a little bit shorter um I'll just say the the first half of the book is um predominantly dedicated to a lot of the more theore iCal aspects of active inference um there will be explanation of other aspects such as the the mathematics involved although you'll find much more on the details of that math in the appendix um the active inference Institute also um if you view the Koda you'll find like references to the maths um you'll see that there's an ongoing project right now and it's being streamed regarding the mathematics of of active inference so there are plenty of resources outside of this textbook um for those who are more interested or or feel like they're not getting a sufficient amount of explanation from um and then we'll also later in this first part of the book we'll also get references to uh the kind of the more neurobiological aspects of the theory of course it was um largely initially devised by Carl friston as a neuroscientist um a neurobiologist and so the all of this is heavily based within preceding work and looking at how to understand the mind and the Brain um so with that said I think beyond that we'll we you know key words include active inference is of process Theory we're looking at different kinds of models discret and continuous models we're getting references to yeah just a variety of things that this is connected to Hybrid models um I think that might be fair enough we're trying to introduce the the chapter we're already nearly a quarter of the way through so um so I'd like to ask if there's any are there any questions that anyone has about the textbook or anything that they've kind of been chewing on for some time you'd like to discuss um so can surprise be seen as the difference between expected observations and actual observations yeah I'm sorry I missed just the first couple words of what you said could you repeat that for me please yeah um so so I was asking um can surprise be seen as the difference between expected observations and actual observations that's right so

so active inference kind of relies upon this um idea that you can think about it as um like a like a set point for example um in in control theory or or looking at a thermostat right so you set the thermostat to a particular temperature um there's a signal that's being read about the temperature in the room um here's what you want the temperature to be here is what it actually is um and then the idea is you'd want a heating or cooling system that could then take action here to bring the the temperature that it's reading in line with the expected set point temperature that you want um and you can quickly see like yeah you're you're minimizing the difference between expected and actual it's a kind of classic machine learning persp perspective or or otherwise um yeah that you'd be effectively minimizing surprise um in that sense but mathema Ma sorry mathematically it's easier to minimize variational freey than it is to try and minimize the area directly right that's right y the textbook will get much more into that but it's it's that itself is also very much based upon mathematics it's not a arbitrary decisions so but it it has to do with Jensen's inequality um the idea that um basically a particular quantity is always necessarily greater than or equal to another one given certain conditions and so that's how we end up there um thank you I ask something uh sorry can you hear me yes yes uh i u a lot of things hinge on the generative model that an agent or an organism has and I I often wonder it like and and yeah when I read further in the book then I see that the generative model is considered given somehow you you you seem to you you seem it's considered to that you have access to it uh so then I wonder where does that model come from who chooses what model is there uh the parameters seem to be updated through message passing and that kind of uh stuff but uh uh the model itself um it's the the P and not the Q but where where does it yeah where does it come from or or how should one think about that before I give an answer I was just wondering if anyone else um wants to chime in on that thought because it is a big question it's um I guess uh does the free under oh I'm sorry people are raising their hands uh let them go first yeah Ali Anna did you have a question oh yeah but Ali was is gonna go first he was first uh okay so what I wanted to comment was a bit unrelated to this current question so Eric would you please uh go on about what you were talking about before thanks oh I I was just going to say um I think is his like how the model is updated is is that um is that really addressed directly by the free energy principle or is it kind of like mysterious right because um it seems like uh you kept on you keep on getting this phrasing from uh friston like this is how a an organism has to behave if you know um so uh um I don't know coming from a neuroscience perspective I'm not sure we quite understand how it works but we could see that the brain seems to be doing it something like this I I don't know if that's

a good answer but yeah know that that makes a lot of sense I mean it's so to I mean come at to come at any kind of and this is just me speaking but to come at any kind of phenomena I presume you have to you know you start with some kind of framework where you attempt to like kind of structure things and then and then as far as living organisms and the behavior goes yeah it's I mean all of this is sort of um you know the map is not the the territory we're just attempting to understand the organisms right and so active inference is like attempting to provide that so the just to comment on how you phrased it as like it's as if uh it is undertaking free energy minimization I think that's a just fine way of saying it um so as far as yeah defining a generative model I suppose it depends on what kind of perspective you want to take so there are many people at the Institute who are more focused on the modeling and and the programming right and so they want to construct like a model themselves U maybe it's to simulate something like an agent it's an environment um in that case yeah you're kind of you would want to rely upon things like domain knowledge in in conjunction with with a lot of the principles of active inference and basing mechanics which as we mentioned earlier is kind of the low road to active inference um that's already your starting point you you already assume that the agent is attempting to minimize free energy and it's doing it through a variety of principles all related back to basis theorem um so that that's kind of the the top- down normative framework that's being used here and then from there it's sort of up to you as far as how you want to continue constructing that model right so if if if if there's certain actions that are available to the model or not that it can take um what are the different kinds of states it already expects to see what are the the priors this this is the kind of language that we'll get more deeply into is the the textbook you know as we continue reading through the chapters um from another perspective if you're not modeling but rather just kind of going through the theory here I mean it's there can be bigger philosophical questions about um the structure of an agent why it is the way it is um I think that there are many people who study active inference who kind of follow the line of of something along the lines of natural selection or perhaps more nuanced um versions of that theory uh you know since its initial conception and so I don't want to say too much here but has that at all been helpful in in trying to answer the initial question yours yeah I I see that this is uh this is not not an easy question and not uh but I I understand that there is for a modeler there is a lot of room for uh you say creative filling in of the structure of the model at least it seems uh while um yeah if I get you right then it's not something that is necessarily derived from uh a normative point of view or something yeah I mean of of course like with any kind of scholarly pursued I mean whenever there's already

work that's been done whenever there's already secondary literature to draw from and so on of course folks will build upon one another's models and test things out and oftentimes you'll construct one model and then eventually realize like it's Its Behavior despite the fact that it's minimizing free energy the way you might have set up the variables in the model like the behavior at elicits doesn't actually match any kind of real world phenomena that you're actually you might be attempting to simulate for example um in that case it's it can be a trial and error process um it can involve discovering something that you didn't expect to find when you realize oh there's a missing variable here um yeah those are many of general uh trials and errors of model construction I suppose um yeah thank you um Anna I did want to come back around to you yeah um okay so actually my [Music] question um is relevant to the previous question I think um so on page six of the book um there's you know this sentence that I'm going to read it says active inference is a normative framework to characterize Bas optimal behavior and cognition in living organisms and so I was wondering if you could um give us an overview or some um ideas to start thinking around the you know what what does a basian model optimize and um so because that's the basian optimization is a tool that allows us to think about the world and so I'm curious about things things like what what utilities are uh optimized in basian models uh is that are the utilities um specific to the phenomena that you're working with um and and if you could connect and discuss you know the utility to the probability functions that are used in basian stats statistics so again this time I do want to open the floor if anyone has any any thoughts on on this or otherwise um to maybe see if we can get into this um this question that an's posed go foron oh well I I had my hand raised from to ask an independent thing but I guess like in terms of um I can try to take a stab at this one um so I would I would phrase it as beian models don't on their own optimize for anything but the model itself States the relationship between variable that is assumed and under those assumptions one can in light of new evidence um Express a posterior distribution which describes one's beliefs and I think classically that's like fully separate from how we think about utility uh utility functions and sort of given my posterior distribution about the world I can then choose to act depending on what my my preferences are can you can you um add to your example like uh a you some you know what what is the currency that's being optimized it just I understand that the modeler decides what currency is being optimized So when you say when you're talking about beliefs updating beliefs and so on what are we optimizing there you know so how do we translate uh the you know the natur the natural language um updating beliefs what does that look like in a in a in a model and can so one one way that I would

anyone feel pry derail to derail me on this uh one way that I think about um ban updating and and I think like the currency that that's at play here is uh units of surprise so bits or Nats right and I think like the the textbook doesn't really talk about this I feel like other sources are better about like grounding bits or kns as like a unit for surprise but like if you were to have a lot of observations from a distribution and you want to communicate those to some other party efficiently um literally how many bits do you need per per observation um that's the kind of thing that uh is reduced as your model of the source of the data is improved so if you're if you're constantly surprised then you're going to take more bits per observation to communicate what you're observing and if your model is pretty well explaining what you're seeing then you'll need many fewer bits per observation to to report or communicate um and I guess maybe communication comes like loaded with some expectations of like between agents but I think also that can be between components within an agent in in some cases but yeah that bits or kns are like actually an expression of uh perplexity surprise um on average I suppose um that can be uh reduced by updating appropriately thank you okay and I I do want to add to that to um I think it was well said Erin in pointing to yeah it's it's frequently understated like what the units of measurement here are whenever we're looking at something like surprisal um that said if we want to look at it in terms of active inference I mean some some things that you're going to see are are commonalities throughout the textbook is we're going to be looking at particular variables and so um we're going to be looking at priors which can be viewed as something like prior predictions what you think is going to happen um or or is happening um I.E your hidden state that you're estimating and then you'll have sensory observations right which is new information that's coming in um this is where we get into things like predictive coding or processing at least uh where there is a continuous process of um comparison between what information you're getting versus what you thought would happen this kind of hearkens back a little bit to the general idea of you know here's the ex expected observations here's the actual ones like the thermostat examples and it's through kind of starting with bases theorem as a sort of foundation like that's the how of of reducing surprise right so um given some new information exactly I mean I don't want to rehash what Aaron said because I think you put it much more eloquently than what I'm prepared to but um yeah it's to um to minimize surprise via uh taking in new information updating what you thought you knew before suddenly your predictions perhaps come more into line with what you see out in the environment it's much more complex than that depending on the situation but um that's an attempt at getting closer to describing you know here's what we're doing whatever War minimizing surprise um Erin I do

uh you've had your hand up for some time so I'd like to allow you to speak on on your original question yeah okay so one thing that I struggle to get my head around here is the generative model um is supposed to generate like all of your sensory data or is like supposed to like go all the way out to like you know the the margins of the mark blanket or whatever the things that are coming in from the exterior World um and how plausible is that really like there's lots of cases where you can see a complicated situation and not be able to predict most of it um and a lot of the things that the details that may be surprising possibly aren't going to be relevant to any action uh you know you look at um a whole bunch of birds flocking together you don't know where any particular one of them is but it kind of doesn't matter uh if you look at a TV screen with a bunch of static you can't predict what color any pixel is going to be in the next millisecond but you can predict like derived features about a scenario um and so I'm wondering sort of there should be some intermediate layer uh that is more relevant for our generative model to populate than like the actual sense data coming in does anyone have any thoughts on that uh actually this is uh directly relevant to John Verve's relevance realization uh because um in the in that framework uh it's not it's not possible for an organism to encode every single bit of information about its environment uh I mean with uh sufficient Precision so uh it needs to only encode uh only the information uh which is relevant uh relevant uh to the uh situation uh which it's dealing uh with with currently so um one way to do that is to uh is to encode those information probabilistically instead of deterministically because uh in uh our traditional algorithmic formulation of cognitivism for example uh we deal with with well-defined situation in which the parameters are pretty much clearcut clearcut and we can encode them uh with u arbitrary Precision if you want but uh in one of the selling points of predictive processing Frameworks is that uh it's not necessary to uh to encode those information with any uh kind of precision at all you can even start from uh I mean in principle you can start from a from an arbitrary Precision from an arbitrary distribution and then uh and then move your way uh in in a in an optimization scenario uh so for in for instance you can do a gradient descent on the VF landscape and reach uh the extrema or the minimum uh of the variation of free energy and uh that's uh one of the one of the fundamental differences between uh active inference or predictive uh processing Frameworks uh from the other traditional uh cognitivist uh Frameworks because uh as this recent uh paper naturalized in relevance realization shows uh in real world real world situations cognition might be fundamentally incomp uncomputable so but um we can uh I mean get around uh this obstacle by uh somehow uh encoding uh that generative model probabilistically uh and only encode the information that are relevant uh to

us yeah I think there there have been arguments that have been made especially just a brief tangent that that you know living organism living organisms who who necessarily like have um kind of metabolic um rules to to keep in place and you know home homeostasis to maintain tendency to we we can view this as one potential explanation for something like only paying attention to what is surprising and meanwhile trying to kind of put on the back end all the processes involved in um you know determining like what is the new information here and what is not the new information do we need to know the position of every single seagull in the flock as it moves by to be able to comprehend that this is a flock of seagulls right um it you know that I I think it just it kind of hits on the aspect of of active inference but it doesn't even necessarily have to be active inference like the importance of something like a having a variational dynamic that allows for things like dimensionality reduction and increase from there compressing information and um just kind of keeping things a little bit more efficient um in any case um I would like to give others a chance to ask questions uh how apologies if I um yeah thanks so that um yes so I guess you know I guess coming from machine learning as well I I try to I guess compartmentalize my my learning here and um you does active influence in influence news um new networks gradient descent um and you like back publication or is it like a just you know different I'll give a quick answer to this there's an essential analytical form that is just matrix multiplication so what we see in the textbook is just matrix multiplication how however a lot of the modern uh examples have used neural networks as kind of like plug-and play for different things like instead of a massive a matrix you can have a neural network and there's also deeper theoretical Connections in live stream 51 with tuya isura this discusses the monotonic relationship between the variational free energy and the neural network uh loss function and then live stream 52 or like anything with Lance D Costa is super informative about how accelerated optimization methods are used with variational free energy some of which are um generally accessible to any variational method others are especially enabled by act in for FP interesting so it can use new networks but it it doesn't need to exactly oh okay um oh I had a question but yeah I've lost it um yeah thanks for the answer by the way uh Thomas yeah I mean just kind I want to build on some of the questions and looking for examples on that generative model point I remember the the frisen article on aesthetic experience uh Order and Chaos there's a note in it about the nature of absences so there's like visuals basically of a square with a piece cut off and there's a comment in the caption they talked about absences are meaningful as obstacles or prediction error so I think that's really interesting because that's pointing out that a kind of absence um it's almost like where's

the grounding like where's the gender of mod come from it seems to be and that relates to the other parts of that essay where he talks about this kind of epistemic promise that our we're using we engage in AR the experiences because there's been a promise that we're going to get some kind of payoff of epistemic basically Clarity or resolution of some kind um there's a lot more Nuance in that but just I want to point that just to kind of build on this point where is this Jen of model gr bottom up thing coming from absences or lacks basically that are pointing out that that's a kind of course basically either there's an objective you're trying to achieve that's being hindered so there's an obstacle in that mode or there's a basically a difference between the the prediction model in your head versus the reality of interfacing with it um I think that's interest that's kind of that was an interesting just building point on where potentially a generative model comes from um also just other idea like where I'm thinking in my head I was looking reading this piece and where this comes from concretely as an example it's like you got a gallon of milk in the fridge you think it's full you're not paying attention so you recruit way too much energy boom and you pull you You Yank it out with way more Force than expected uh that difference basically is a surprise I mean that's why I'm thinking as a concrete example of um so in that case what's the currency being manipulated years is calories basically to bur basically to recruit the energy required for the muscles to contract and move on that example those things I wanted to share yeah no good good points than for sharing those I mean yeah again with the you know expecting a much you know jug of of milk is another very quick intuitive example of you know here's the difference between what you predict is is currently going on the hidden state of the world like I have a full jug of milk um what actually happens there's much less uh milk in there you know you end up like yanking it out way too quickly there's excessive metabolic like expenditure there um those are all ways of for sure thinking about it and and you know uh at some point in the textbook we'll see explanations of things like uh Arc reflexes um and proprioception where uh it'll describe things like like for example you can move on from that example you gave to imagining like the arm will adjust right like well we hope it does anyway like the arm will not just immediately fling out it will like maybe halfway through a quarter of the way through the movement start to pick up like oh this is actually not heavy enough to justify this amount of force there'll be some kind of change in the action sequence right like suddenly you're not hopefully you're no longer G to fling it sorry it's kind of a silly example isn't it but um but you see the point it's like error correction right so so there's what you expect there's what's you're actually observing and then there's an um action can then be used to try and correct for that error um which ultimately uh

presumably minimizes free energy along the way Eric um yes I was just gonna also add um to the question raised earlier that maybe part of the answer to the question about whether um the model really has to um model every Century detail is just that there's there 's actually many models that are working together um in in in an organic system so um they don't all have to model everything um and uh like if I understand uh B's take um I think he believes that like the um task task centered Network and the default mode Network which kind of take turns right when you're um when you're doing various activities that you know the um the uh default default mode network is basically aspectual using different things in the world and um like offering them up to the um task Focus Network which is selecting some of them um and that they're constantly mutually modeling each other so that the organism can move through a salience landscape but um you know um we tend to reduce organisms you know being conscious creatures we tend to think of the model as oh it's me the conscious me right but there's all these um my understanding is that there's all these um generative models right working beneath my Consciousness um allowing some things to appear in others you know um so maybe that's part of um how it works efficiently because there's many different models hopefully that's coherent yeah know it makes a lot of sense to me and I I think um you know this where we start to see a little bit more relationship with um yeah as you described raiki theories as well as you know if anyone's familiar with like Global workspace Theory basically this the idea that you can have a kind of uh like multiple systems within a single system or nested systems and um we're already in chapter one briefly introduced to the idea of a marov blanket but um as the textbook continues there'll be some more regarding Markov blankets and what what separates statist in this sense stat also statistically separates a system from another system and then also recognizing that there can be like nested systems within systems or systems who appear to be working cooperatively together and so in the context of like a human being um we only need a like a you know what's a quick example someone who uh is a cigarette smoker or something right like someone who on the one hand like and decides to quit that's sorry I'll keep it more brief than that um someone who decides they want to quit like top level hierarchical like higher order decision I want to quit I want to be done with this um this is good like that's what I want to do that's a higher order decision on the other hand uh you know give them you know roughly 15 minutes and they start experiencing a craving and as that time continues we have you know withdrawal symptoms and so on um suddenly the person will become potentially more ambivalent um there are different systems in the body that are expecting that nicotine uh to to come in right um and so you can have like multiple

systems within a single human being or a single system that are potentially in conflict with one another cooperating with one another um it's really about what is the phenomena you're interested in whenever it comes to active inference and applying it as a framework what is the what is the system of for you and and you have to decide like what defines those boundaries like what Behavior am I expecting to see what system am I trying to to um simulate here what potential nested systems are within that broader system um so at risk of over speaking I'll I'll uh give the floor back but uh no I I think those are all great questions and uh things to think about Ali yeah I just wanted to add uh to what Thomas just mentioned about um I mean uh the the absence of information uh because that's exactly uh the premise of U Terence Deacon's incomplete nature because uh in his magnumopus he develops theory of absentism and he tries to somehow uh derive kind of Dynamics based exactly on those uh what he calls appential so uh in one of our works with uh ma and I U which is INF interaction inference uh we uh we initially developed an account of the interaction between the organism and the environment based on possibility spaces so uh very briefly any system with a given State space uh would have a form of the possibility space based on uh the definition or the collection of uh its Tendencies and capacities and uh and the structure of this possibility space can be described uh or defined as a multiplicity as a delusion uh multiplicity so by multiplicity uh what we mean is a nested set of vector Fields uh which are related uh to each other hierarchically by symmetry breaking bifurcations uh and uh which together they constitute a kind of a collection of distributions of attractors um uh which Define each of the uh their success their subsequent or embedded levels so in other words uh the idea of essentialism is already um embedded in these uh spaces of possibility ities or the multiplicities because those are exactly what constitute the structure of uh the state of the space of possibility as uh as the attractors of the multiplicities so uh in one of our recent papers we we're trying to develop uh this connection um a bit more explicitly and uh yes uh I agree that um those appential are uh quite essential for uh any kind of representational framework uh or generative models so I have a question Ali about absentism and sure uh yeah so how can I ask this um so in Bas in statistics we are using probability distributions to get at understanding something about reality um can you define entails in the context of probability Theory or is it completely unrelated okay so uh first of all as um as a side note uh Terrence Deacon actually uh doesn't use the term absentism in his book and uh he even rejects uh this word explicitly but um in any case in all of the literature related to incomplete nature they've used the word absentism so uh it's pretty much um widely accepted so uh by absentism he means something that's not uh

physically realized yet or in other words uh as would call it it's not actual but a virtual possibility of uh the system so uh for instance as a very simple example you can think of the capacity of a knife uh to cut to kill or I don't know to to poke something so uh so yeah all of these are kind of capacities of the the of the knife uh which are not always actualized they reside as uh as its virtualities uh in its space of possibilities right and uh the knife may or may not ever actualize these virtualities but it uh it doesn't make it any less real so by appssential absenti Concepts um or as another Rel term of Deacon um intentional Concepts he means something similar uh kind of possibility or potentiality uh which is not actualized or realized yet uh but uh it is real and it constitute uh I mean the uh the D the internal dynamics of the system uh as um as its potential as its capacities and Tendencies so uh I'm not sure if we can characterize uh absenti ISM in terms of probabilities because uh although it it can be said that some of those virtualities uh are more probable than others to be actualized uh but uh but it needs um a much more rigorous uh work on that uh to uh to derive these kinds of probabilistic landscape over those multiplicities but in any case that multiplicity exists as a kind of reality of the system or the agent uh again whether uh I mean regardless of U whether it actualizes those possibilities or virtualities or not it would would it be incorrect to say that there are states that are virtually or have a potential that they exist in some virtual representation somewhere and and that those states are not uh accessible to us for whatever reason for to The Observer at some point in time but yes to me to me that that translates let's see I'm trying to see if I this is making any sense to me that translates into in order for me to even think about that thing that's absent I The Observer need to have some sense of what what could possibly what could possibly be absent so I as the Observer have to also have a virtual version of the thing that's absent and exactly and would and would that thing that's absent not have a probability distribution so you see uh these appssential kind of reside uh in the space of possibility for this the space of possibility of the system right uh but uh in order to actualize those possibilities of course they can be prioritized based on the probabilities of U I mean those absenti to be actualized as I mentioned earlier some of those virtualities uh can occur with a higher frequency than others right uh but this is this is a very subtle point and active inference uh which is to say that uh you see in recent years I mean uh beginning from 2022 uh there's this uh attempt to somehow um blur the line between uh The Observer and The observed by showing a complete Symmetry and Duality between Fe and cmap uh which is the constraint maximum entropy principle right so in other words uh what the the recent formalism of active inference or rather FP shows is this complete Symmetry and Duality

between uh these two perceptions these two I mean distinctions between observed and the Observer so uh every time we uh draw a mark of blanket a statistical boundary between a system or between the Observer and The observed it can basically be treated as a kind of arbitrary boundary we can uh we can switch that observed uh and observer perspective um and uh the formalism uh will help perfectly so uh um yes there is this Duality and perfect symmetry between Observer and observed which is shown uh formally in recent work by Dalton Sac divided El and others and especially from the Viewpoint of category theoretical perspective it can be um shown much more eloquently uh through those um category theoretical formulation as well such as the work of um uh Toby SMI and others thank you appreciate it um unfortunately I see we're just about out of time here it's been we' run through the the whole hour and for a I will say for a discussion on chapter one of a textbook this got rather deep uh very interesting stuff though very happy to hear everyone share their their thoughts and uh I would you know of course everyone here is coming from their own distinct perspective and and interests so I would say you know if you are coming from a certain angle certainly don't hesitate to to ask questions I think that's what'll well what will kind of pull you into to active inference a little bit more quickly in a way that you know is a little bit more fluent um for a lot of people because it can be a lot of material at one time but um thanks Ali for for Co facilitating and and your very U your very thorough responses I always end up learning something new about uh active inferences history and and sub theories that makes me want to read more um and uh thanks everyone else uh for attending uh per usual there will be another meeting next week uh we'll be going over chapter two which is where we'll get into the low road to active inference a lot more on basian statistics and mechanics um we'll get a maybe a clearer view of what optimality means or optimal uh behavior and inference um we'll also get a couple perspectives on things like uh different time scales such as inference at a very quick Moment by moment scale all the way up to uh planning and agents who actually have the capacity for planning their next move or or or constructing you know goals that will take a series of steps to carry out not just taking a breath in the moment because you need it um Daniel do you have anything to to add or announcements no great all right thank you everyone farewell thanks every thanks everyone bye